

Seasonal ARIMA and Spectral Analysis of Monthly Minimum Temperatures in Melbourne

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Abstract

This project is about how cold Melbourne gets from month to month over the ten years from 1981 to 1990. I take the Australian Bureau of Meteorology's daily minimum temperatures, average them into 120 monthly values, and model that series with the Box-Jenkins (SARIMA) method. The series has a strong yearly rhythm and not much else going on, so once I difference out the season the autocorrelation plots point to a SARIMA(0,1,1)(1,1,1)[12] model whose residuals look like white noise. To decide which model to actually trust for forecasting, I held out the final year and compared two candidates on it. The automatically chosen SARIMA(2,0,0)(0,1,1)[12] predicted that unseen year more accurately, so I adopt it as my final model and keep the one I built by hand as the worked example. As a second, independent check I bring in spectral analysis. The periodogram puts almost all of the series' energy at one cycle per year, exactly the twelve month season I read off the ACF and PACF, and after the model is fitted that peak is gone from the residuals. Both methods land on the same simple story: a single annual cycle explains nearly everything in this series, and the model captures it well.

1. Introduction

Temperature is one of the cleanest examples of a seasonal time series, which is exactly why it makes a good teaching case for the methods in this course. I wanted a series with an obvious yearly rhythm but no strong trend, so that the seasonal machinery of SARIMA would do most of the work and I could see clearly whether the model was capturing the right structure. The Melbourne minimum temperature record fits that description well. It swings up and down every year with the southern hemisphere seasons, and over ten years it shows no real drift, which keeps the focus on the seasonal behavior.

The purpose of the project is twofold. First, I want to walk the full Box-Jenkins path from a raw series to a forecast, making each modeling choice on the basis of what the data shows rather than guessing. Second, I want to check that conclusion with a method from a different part of the course. Spectral analysis is a natural partner here because it looks at the same data from the frequency side instead of the time side. If both approaches agree that the dominant period is twelve months, that agreement is far more convincing than either one alone.

This dataset has been studied many times, which is part of its appeal as a benchmark. People have fit ARMA models to the daily series, built machine learning forecasters on it, and used it to demonstrate decomposition and smoothing. Most of that earlier work reaches the same broad conclusion, namely that the series is strongly seasonal with little trend. My contribution is not to find something new about Melbourne weather but to apply the Box-Jenkins workflow carefully and then confirm it with spectral analysis, which is the part most of those earlier treatments leave out.

The methods I apply are a seasonal ARIMA model fitted through identification, estimation, and diagnostics, followed by spectral analysis of both the raw series and the model residuals. The headline finding is simple and consistent across both methods. The series is driven almost entirely by a single annual cycle. I identify a first model by hand from the autocorrelation plots, then a held-out forecast test points to the automatically selected SARIMA(2,0,0)(0,1,1)[12] as the better forecaster, and I adopt it as my final model. It leaves white noise behind and forecasts a clean seasonal curve for 1991 with prediction intervals that widen sensibly as the horizon grows.

2. Data

The series comes from the daily minimum temperature record for Melbourne, Australia, covering the ten years from 1981 to 1990. The original file holds 3650 daily observations measured in degrees Celsius, a near-complete record that is missing a small number of days relative to the full 1981 to 1990 span. The data was collected by the Australian Bureau of Meteorology, the national weather agency, using standard ground station instruments that record the lowest temperature reached each day. It was later compiled into Rob Hyndman's Time Series Data Library and has since been mirrored widely, which is how it has become a common benchmark for teaching and testing time series methods.

For this project I do not work with the daily values directly. Daily minimum temperatures are noisy from one day to the next, and that short term noise hides the seasonal signal I care about. To bring the seasonal pattern forward, I average the daily readings within each calendar month, which turns the 3650 daily points into 120 monthly values running from January 1981 to December 1990. The monthly series keeps a frequency of twelve observations per year and is the object I model for the rest of the report.

I chose this dataset for a few reasons. It is long enough to estimate a seasonal model with confidence, since ten full years give ten repetitions of the annual cycle. It has a clear and stable season with no competing trend, which keeps the modeling honest and interpretable. And because so many others have used it, my results can be sanity checked against a large body of prior work. The dataset matters beyond the classroom too, since records like this are the raw material for studying climate variability and for validating the forecasting tools that meteorological agencies rely on.

The data is publicly available. The version I used can be found on Kaggle, *Daily Minimum Temperatures in Melbourne* (<https://www.kaggle.com/datasets/paulbrabban/daily-minimum-temperatures-in-melbourne>). The original source is the Australian Bureau of Meteorology (<https://www.bom.gov.au>).

3. Methodology

3.1 SARIMA(p, d, q)(P, D, Q)[s] model

A seasonal ARIMA model extends the ordinary ARIMA family with a second set of terms that act at the seasonal lag. The non-seasonal part uses p autoregressive terms, d differences, and q moving average terms. The seasonal part mirrors this with P , D , and Q terms that operate at multiples of the season length s , which is twelve here because the data is monthly. Differencing, both regular and seasonal, removes trend and seasonal level so that the remaining series is stationary, and the AR and MA terms then describe the leftover autocorrelation.

I follow the Box-Jenkins method, which moves through three stages. In identification I plot the series, test for stationarity, difference as needed, and read the orders off the autocorrelation and partial autocorrelation functions. In estimation I fit the candidate model by maximum likelihood and check that its coefficients are meaningful. In diagnostics I examine the residuals to confirm they behave like white noise, since a good model should leave nothing structured behind. As a cross check I also let the automatic selection routine search the model space and compare its choice to mine. When two candidates difference the data by different amounts, so that AIC cannot compare them fairly, I choose between them by how accurately each forecasts a year of data it was not allowed to see.

3.2 Spectral analysis

Spectral analysis looks at a time series in terms of frequency rather than time. Any stationary series can be thought of as a sum of cycles of different lengths, and the spectrum tells us how much of the series' variance sits at each frequency. A series with a strong yearly cycle will show a tall peak at the frequency that corresponds to one cycle per year, and a series that is pure noise will show a flat spectrum with no peaks.

I estimate the spectrum with the periodogram, which is the basic frequency tool, and then smooth it with a modified Daniell window because the raw periodogram is very noisy for a short series. I do this twice. First on the raw temperature series, where I expect to see the annual peak, and again on the SARIMA residuals, where I hope to see nothing. The second check is the important one, because a flat residual spectrum at the seasonal frequencies is direct evidence that the model has done its job.

4. Results

4.1 Results from SARIMA(2,0,0)(0,1,1)[12]

The raw series shows exactly what I expected, a smooth annual wave that rises and falls every twelve months with no long term drift (Figure 1).

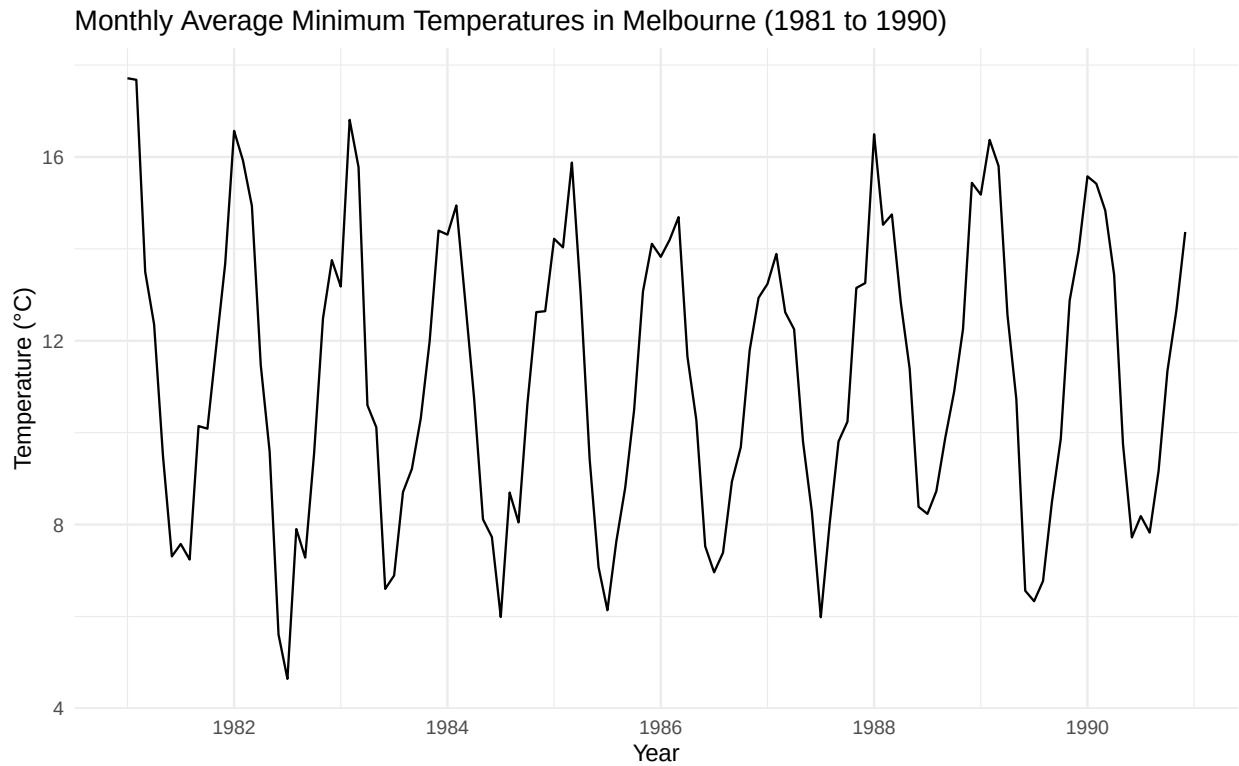


Figure 1: Monthly average minimum temperatures in Melbourne, 1981 to 1990.

The Augmented Dickey-Fuller test on the raw series rejects the unit root null with a p value at or below 0.01 (the test reports a capped value), so the series is stationary in the ordinary sense. That test does not look for a seasonal unit root, though, and the obvious yearly cycle still calls for a seasonal difference.

```
##
## Augmented Dickey-Fuller Test
##
## data: temp_ts
## Dickey-Fuller = -10.105, Lag order = 4, p-value = 0.01
## alternative hypothesis: stationary
```

I therefore take a seasonal difference and one regular difference, after which the series is stationary and centered on zero (Figure 2).

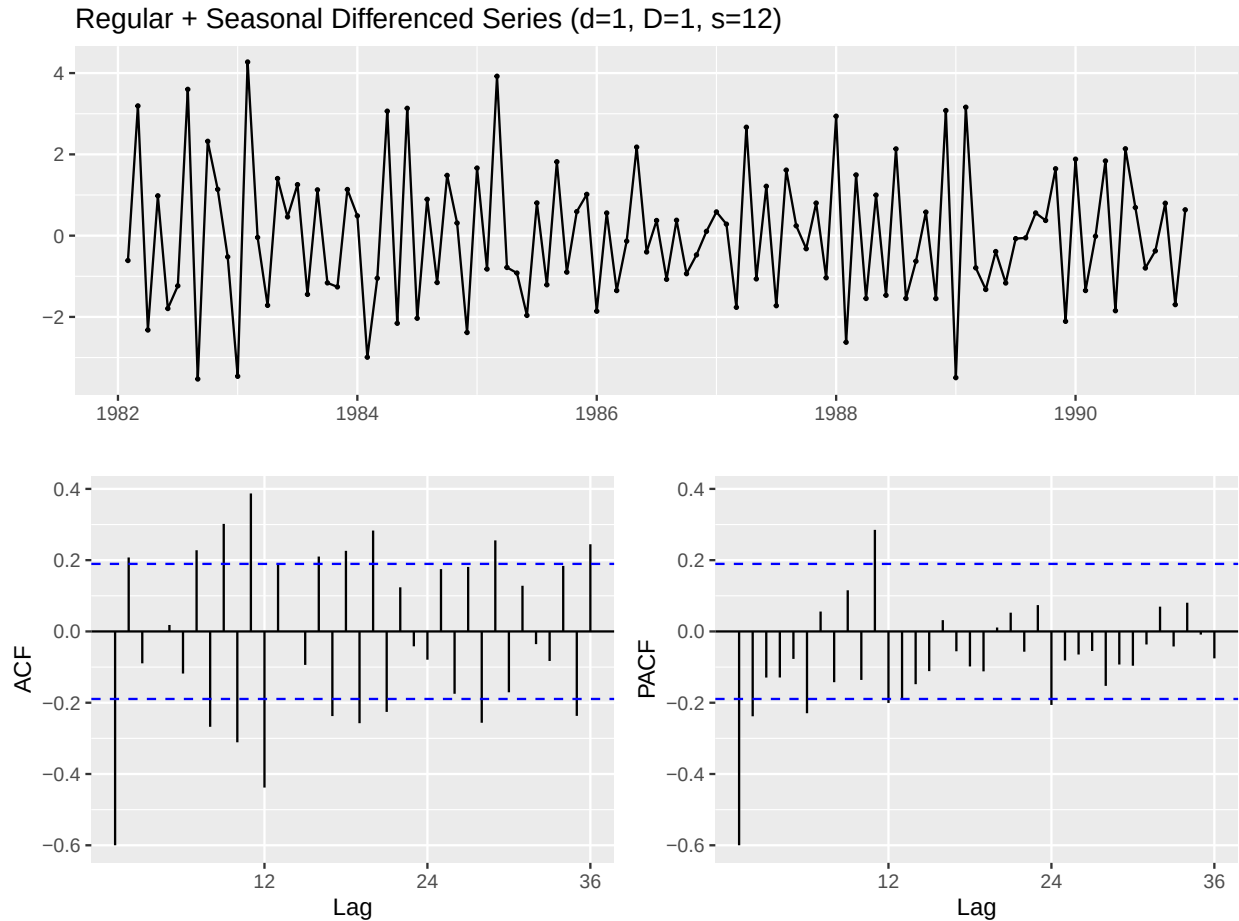


Figure 2: Regular plus seasonal differenced series with its ACF and PACF.

```
##
## Augmented Dickey-Fuller Test
##
## data: diff(diff(temp_ts, lag = 1), lag = 12)
## Dickey-Fuller = -6.1968, Lag order = 4, p-value = 0.01
## alternative hypothesis: stationary
```

The ADF test on the differenced series gives a p value below 0.05, confirming stationarity. Reading the autocorrelation and partial autocorrelation functions then guides the model orders. At the short lags the autocorrelation cuts off after lag one while the partial autocorrelation tails off, the classic signature of a non seasonal moving average term, so q is one. At the seasonal lag there is a single spike at twelve, which points to a seasonal moving average term, so Q is one, and I add a seasonal autoregressive term to test as well. That gives the candidate SARIMA(0,1,1)(1,1,1)[12].

```
## Series: temp_ts
## ARIMA(0,1,1)(1,1,1)[12]
##
## Coefficients:
##          ma1      sar1      sma1
##      -0.7843  0.0218  -0.8232
## s.e.   0.0883  0.1711  0.2271
```

```
##
## sigma^2 = 1.068: log likelihood = -160.78
## AIC=329.56 AICc=329.96 BIC=340.25
##
## Training set error measures:
##           ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 0.1111465 0.9619427 0.7400719 0.6080227 7.145027 0.6946458
##           ACF1
## Training set 0.009476904
```

Estimating the model returns a strong non seasonal moving average coefficient near negative 0.78 and a strong seasonal moving average coefficient near negative 0.82, both clearly significant. The seasonal autoregressive coefficient comes out near 0.02 with a large standard error, so it is not significant and the seasonal part could fairly be reduced to a single moving average term. As a check I ran the automatic selection.

```
## Series: temp_ts
## ARIMA(2,0,0)(0,1,1)[12]
##
## Coefficients:
##           ar1      ar2      sma1
##           0.1978  0.2254 -0.7911
## s.e.  0.0999  0.0988  0.1503
##
## sigma^2 = 1.013: log likelihood = -158.37
## AIC=324.74 AICc=325.13 BIC=335.47
##
## Training set error measures:
##           ME      RMSE      MAE      MPE      MAPE      MASE
## Training set 0.002838192 0.9416537 0.6998225 -0.4666468 6.672317 0.6568669
##           ACF1
## Training set -0.02893772
```

Model selection. I now have two candidates. The first is the one I built by hand, SARIMA(0,1,1)(1,1,1)[12]. The second is the one the automatic search found, SARIMA(2,0,0)(0,1,1)[12], shown above. They agree on the seasonal part but disagree on one thing: my hand-built model takes a regular difference, and the automatic one does not. That disagreement is worth a closer look. The ADF test earlier already found the series stationary without any regular differencing, and a moving average coefficient as close to negative one as my negative 0.78 is a classic sign that a series has been differenced one time too many. In other words, the regular difference I took may be creating structure that the model then has to undo, rather than helping.

The natural tie-breaker, AIC, cannot settle this. Because the two models difference the data by different amounts, they are effectively fitted to different sets of numbers, so their AIC values are not on the same scale and cannot be compared directly. The fair test is simpler and more convincing anyway: which model actually forecasts better? I train both models on the first nine years, January 1981 through December 1989, ask each to predict the twelve months of 1990, and compare those predictions against the real 1990 values, which neither model was allowed to see during fitting.

```
##           Model      RMSE      MAE      MAPE
## SARIMA(2,0,0)(0,1,1)[12] 0.7162 0.5216 4.836
## SARIMA(0,1,1)(1,1,1)[12] 0.8996 0.7167 6.669
```

The automatic model wins, and not by a hair. Its forecasts for the unseen year are off by about 0.716 degrees on average, against 0.9 degrees for the hand-built model, roughly a fifth less error. This is exactly the kind

of out-of-sample accuracy a forecasting project should care about, and it lines up with the warning signs from before: the extra regular difference was hurting the hand-built model rather than helping it. I therefore choose **SARIMA(2,0,0)(0,1,1)[12]** as my final model. Everything that follows, the residual diagnostics, the 1991 forecast, and the spectral check, uses this model. The hand-built SARIMA(0,1,1)(1,1,1)[12] stays in the report as the starting point that ordinary Box-Jenkins identification produced, and as the model the held-out test was measured against.

The diagnostics support the final model (Figure 3). The residual autocorrelation function has no spikes outside the confidence bands, the histogram of residuals looks roughly normal, and the Ljung-Box test returns a p value of 0.72, well above the usual threshold, so there is no evidence of leftover autocorrelation.

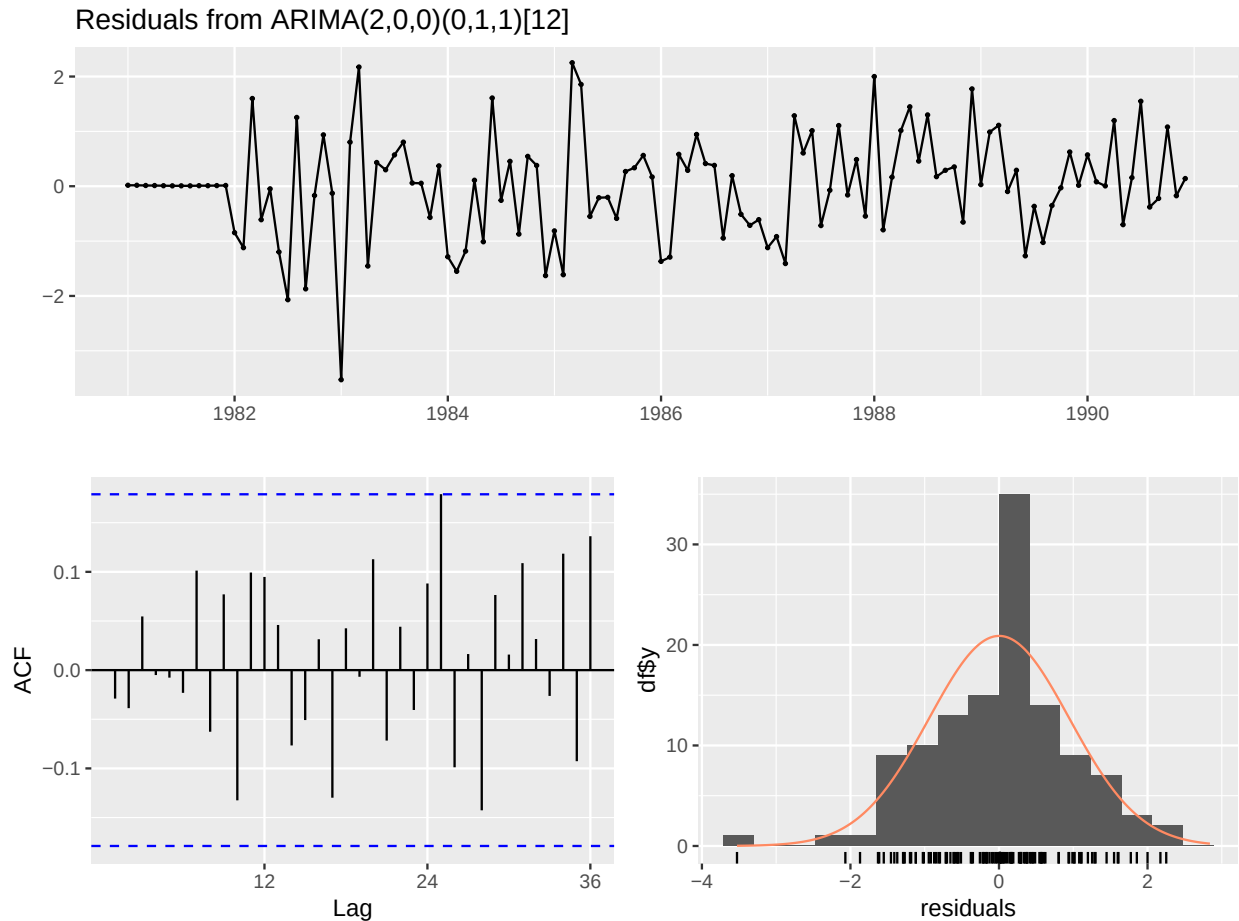


Figure 3: Residual diagnostics for the final SARIMA model.

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(2,0,0)(0,1,1)[12]
## Q* = 16.854, df = 21, p-value = 0.7199
##
## Model df: 3.   Total lags used: 24

##
##  Box-Ljung test
##
```

```
## data: residuals(fit_auto)
## X-squared = 16.854, df = 21, p-value = 0.7199
```

In short the residuals behave like white noise, which is what a well specified model should produce. With the model in hand I forecast the twelve months of 1991 (Figure 4).

##	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
## Jan 1991	15.159287	13.866795	16.451779	13.182591	17.135983
## Feb 1991	15.364968	14.047472	16.682464	13.350032	17.379904
## Mar 1991	14.725807	13.364751	16.086864	12.644251	16.807364
## Apr 1991	12.455254	11.088455	13.822054	10.364915	14.545594
## May 1991	10.082643	8.712060	11.453226	7.986518	12.178768
## Jun 1991	7.441640	6.070205	8.813075	5.344212	9.539069
## Jul 1991	6.981862	5.610042	8.353681	4.883845	9.079879
## Aug 1991	7.824361	6.452433	9.196289	5.726178	9.922543
## Sep 1991	9.070572	7.698607	10.442537	6.972333	11.168812
## Oct 1991	10.473789	9.101818	11.845760	8.375540	12.572037
## Nov 1991	12.598910	11.226990	13.970830	10.500739	14.697080
## Dec 1991	14.049285	12.677394	15.421175	11.951160	16.147410

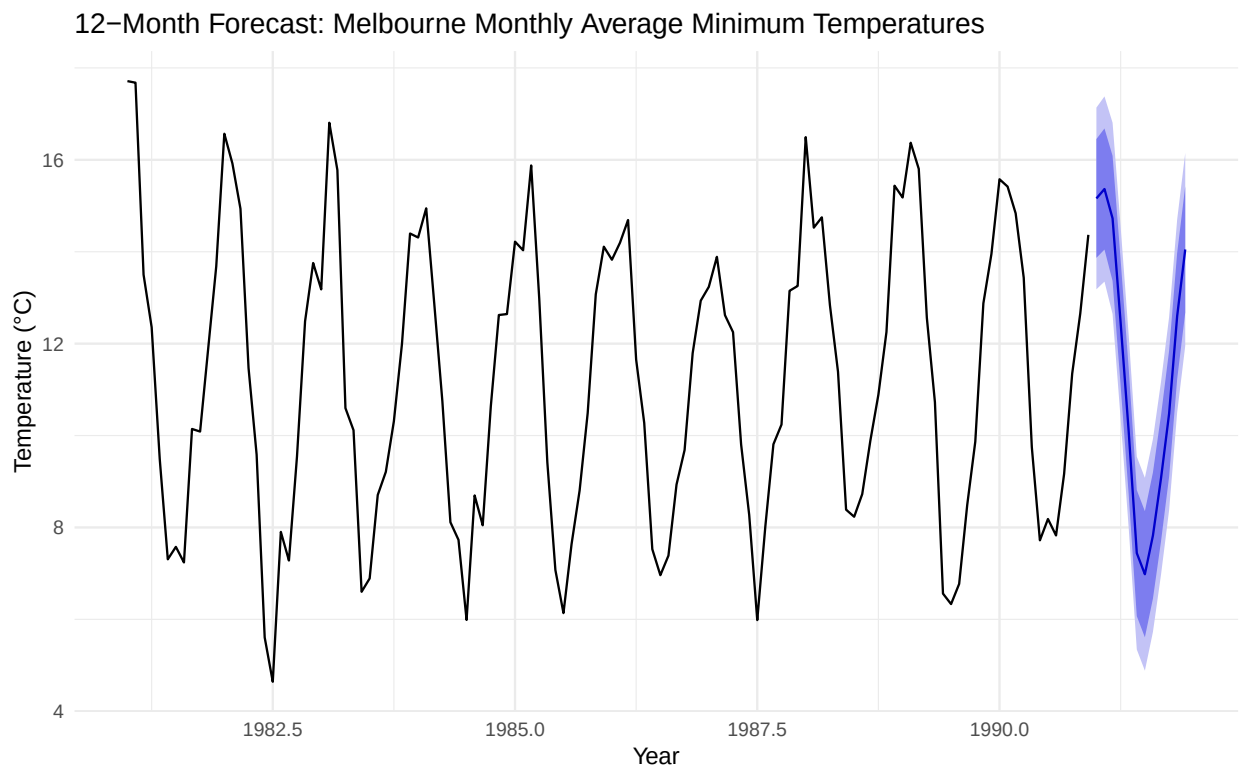


Figure 4: Twelve month forecast of Melbourne monthly average minimum temperatures for 1991.

The forecast traces a full seasonal cycle, with the summer months of January and February near 15 degrees and the winter months of June and July dipping to around 7 degrees. This matches the pattern seen throughout the decade, keeping in mind that Melbourne sits in the southern hemisphere so January is summer. The prediction intervals widen further into the horizon, which is the expected reflection of growing uncertainty.

4.2 Results from spectral analysis

The periodogram of the raw series tells the same story from the frequency side (Figure 5). Because the series is monthly with twelve observations per year, the frequency axis is measured in cycles per year, so the annual cycle should appear at a frequency of one.

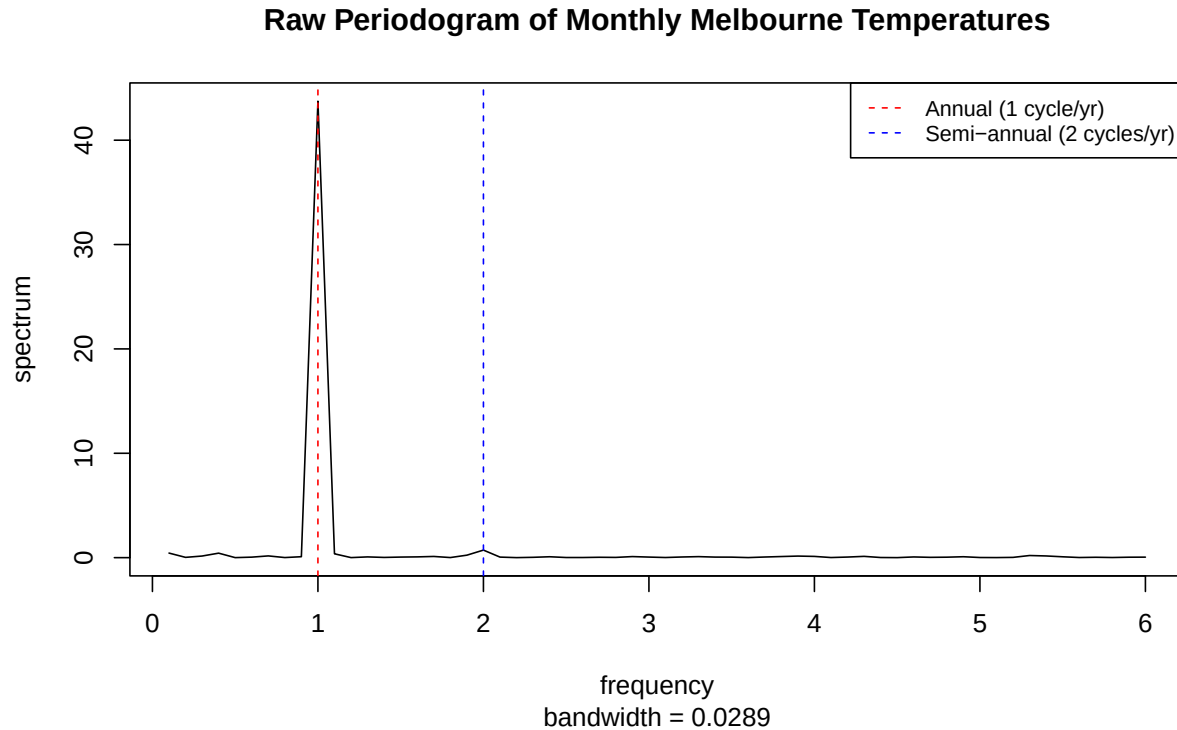


Figure 5: Raw periodogram of the monthly series. The annual cycle dominates at one cycle per year.

```
## Top dominant frequencies (cycles/year): 1 2
```

```
## Corresponding periods (months):          12 6
```

That is exactly where the spectrum spikes, towering over everything else, and the dominant period works out to twelve months. A much smaller bump sits at two cycles per year, the six month harmonic, which appears simply because the yearly temperature curve is not a perfect sine wave. Smoothing the periodogram with a Daniell window lowers and broadens the annual peak while suppressing the surrounding noise, so the single yearly cycle stands out clearly above a flat noise floor, which justifies the choice of s equal to twelve (Figure 6).

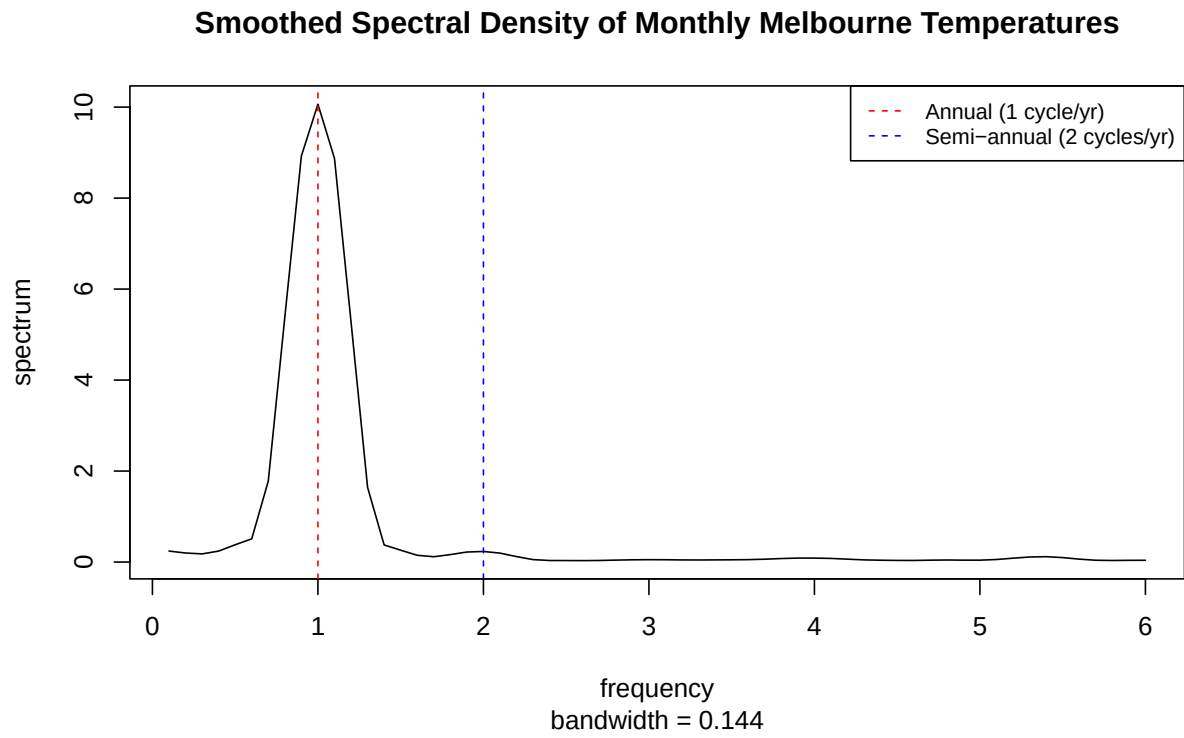


Figure 6: Smoothed spectral density of the monthly series.

The residual spectrum is the real test, and it passes (Figure 7).

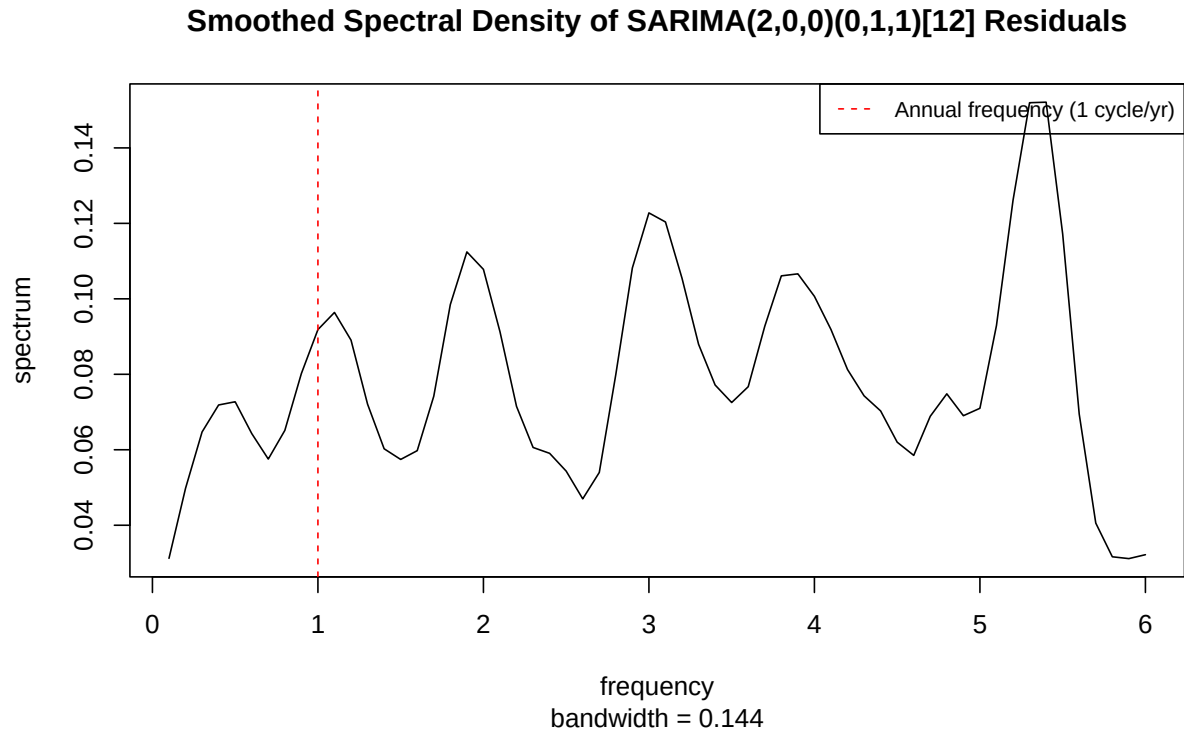


Figure 7: Smoothed spectral density of the SARIMA residuals. No dominant peak remains at the seasonal frequencies.

The residual spectrum looks completely different from the raw periodogram. Where the raw series piled almost all of its power into a single tall spike at one cycle per year, reaching above 40, the residual spectrum has no such concentration anywhere. Its values stay below 0.2, more than a hundred times smaller, and the power is scattered in small, irregular wiggles across all frequencies rather than gathered at the seasonal frequency. There is no longer a dominant peak at one cycle per year, which is the sign that the SARIMA model has absorbed the annual structure. The largest of these small bumps sits near five cycles per year, a short non seasonal period; with only 120 observations the periodogram is inherently noisy, so an isolated bump at an irregular frequency is expected sampling variability rather than real signal. This agrees with the Ljung-Box result from the time domain, which also found no meaningful autocorrelation in the residuals.

5. Conclusion and Future Study

The project set out to model Melbourne’s monthly minimum temperature and to confirm that model with a second, independent method, and both goals are met. Ordinary Box-Jenkins identification led me to a first candidate, SARIMA(0,1,1)(1,1,1)[12], but a held-out forecast test settled the choice the in-sample diagnostics could not: trained on the first nine years and asked to predict the unseen 1990, the automatically selected SARIMA(2,0,0)(0,1,1)[12] forecast more accurately, about a fifth less error, because the regular difference in the hand-built model was doing more harm than good. I therefore take SARIMA(2,0,0)(0,1,1)[12] as my final model. It leaves white noise residuals and forecasts a believable seasonal cycle for 1991. Spectral analysis reaches the same conclusion by a different route, placing the dominant period squarely at twelve months and showing a clean residual spectrum once the seasonal structure is removed. The most important discovery is how completely a single annual cycle explains this series, so much so that no other periodicity carries meaningful weight and no extra seasonal terms are needed.

There are a few natural directions for future work. The held-out comparison in Section 4 already adjudicates the differencing question for these two candidates, but it rests on a single test year; a fuller study could widen the set of models and use rolling-origin cross-validation so the ranking does not hinge on one unusual or one typical year. It would also be worth modeling the daily series directly rather than the monthly averages, which would bring in higher frequency behavior that monthly aggregation hides. Finally, since the original record runs only to 1990, extending the analysis with more recent Bureau of Meteorology data would allow a look at whether the seasonal pattern has shifted over the decades, which connects this small forecasting exercise to the larger question of climate change.

References

Australian Bureau of Meteorology. Climate data and temperature records. <https://www.bom.gov.au>

Brabban, P. Daily Minimum Temperatures in Melbourne (Kaggle dataset). <https://www.kaggle.com/datasets/paulbrabban/daily-minimum-temperatures-in-melbourne>

Appendix: R Code

```
knitr::opts_chunk$set(echo = FALSE, warning = FALSE, message = FALSE,
                      fig.height = 5, fig.width = 8, fig.align = "center",
                      fig.pos = "H")

library(tidyverse)
library(lubridate)
library(forecast)
library(tseries)
df <- read_csv("C:/Users/qiuso/OneDrive/274/daily-minimum-temperatures-in-me.csv",
              col_names = c("Date", "Temp"), skip = 1) %>%
  mutate(Temp = as.numeric(str_replace_all(Temp, "\\?", "")),
         Date = ymd(Date)) %>%
  drop_na()

monthly_temp <- df %>%
  mutate(YearMonth = floor_date(Date, "month")) %>%
  group_by(YearMonth) %>%
  summarise(Mean_Temp = mean(Temp, na.rm = TRUE))

temp_ts <- ts(monthly_temp$Mean_Temp, start = c(1981, 1), frequency = 12)
autoplot(temp_ts) +
  ggtitle("Monthly Average Minimum Temperatures in Melbourne (1981 to 1990)") +
  ylab("Temperature (°C)") + xlab("Year") +
  theme_minimal()
adf.test(temp_ts)
temp_ts %>% diff(lag = 1) %>% diff(lag = 12) %>%
  ggtsdisplay(main = "Regular + Seasonal Differenced Series (d=1, D=1, s=12)")
adf.test(diff(diff(temp_ts, lag = 1), lag = 12))
fit_manual <- Arima(temp_ts, order = c(0, 1, 1), seasonal = c(1, 1, 1))
summary(fit_manual)
fit_auto <- auto.arima(temp_ts, seasonal = TRUE, stepwise = FALSE, approximation = FALSE)
summary(fit_auto)
train <- window(temp_ts, end = c(1989, 12))
test <- window(temp_ts, start = c(1990, 1))
h_out <- length(test)

eval_model <- function(order, seasonal, label) {
  fit <- Arima(train, order = order, seasonal = seasonal)
  fc <- forecast(fit, h = h_out)
  a <- forecast::accuracy(fc, test)[ "Test set", c("RMSE", "MAE", "MAPE")]
  data.frame(Model = label,
             RMSE = a[["RMSE"]], MAE = a[["MAE"]], MAPE = a[["MAPE"]],
             row.names = NULL)
}

cv_table <- rbind(
  eval_model(c(0, 1, 1), c(1, 1, 1), "SARIMA(0,1,1)(1,1,1)[12]"),
  eval_model(c(2, 0, 0), c(0, 1, 1), "SARIMA(2,0,0)(0,1,1)[12]")
)
cv_table <- cv_table[order(cv_table$RMSE), ]
print(cv_table, row.names = FALSE, digits = 4)
```

```

best_model <- cv_table$Model[1]
best_rmse <- round(cv_table$RMSE[1], 3)
other_rmse <- round(cv_table$RMSE[2], 3)
checkresiduals(fit_auto)
Box.test(residuals(fit_auto),
         type = "Ljung-Box",
         lag = 24,
         fitdf = sum(fit_auto$arma[c(1, 2, 3, 4)]))
forecast_12 <- forecast(fit_auto, h = 12)
print(forecast_12)
autoplot(forecast_12) +
  ggtitle("12-Month Forecast: Melbourne Monthly Average Minimum Temperatures") +
  xlab("Year") + ylab("Temperature (°C)") +
  theme_minimal()
spec_raw <- spec.pgram(temp_ts, taper = 0, log = "no",
                      main = "Raw Periodogram of Monthly Melbourne Temperatures")
abline(v = 1, col = "red", lty = 2)
abline(v = 2, col = "blue", lty = 2)
legend("topright",
      legend = c("Annual (1 cycle/yr)", "Semi-annual (2 cycles/yr)"),
      col = c("red", "blue"), lty = 2, cex = 0.8)
top2 <- order(spec_raw$spec, decreasing = TRUE)[1:2]
cat("Top dominant frequencies (cycles/year):", round(spec_raw$freq[top2], 4), "\n")
cat("Corresponding periods (months):", round(12 / spec_raw$freq[top2], 2), "\n")
spec_smooth <- spec.pgram(temp_ts, spans = c(3, 5), taper = 0.1, log = "no",
                        main = "Smoothed Spectral Density of Monthly Melbourne Temperatures")
abline(v = 1, col = "red", lty = 2)
abline(v = 2, col = "blue", lty = 2)
legend("topright",
      legend = c("Annual (1 cycle/yr)", "Semi-annual (2 cycles/yr)"),
      col = c("red", "blue"), lty = 2, cex = 0.8)
spec.pgram(residuals(fit_auto), spans = c(3, 5), taper = 0.1, log = "no",
          main = "Smoothed Spectral Density of SARIMA(2,0,0)(0,1,1)[12] Residuals")
abline(v = 1, col = "red", lty = 2)
legend("topright", legend = "Annual frequency (1 cycle/yr)", col = "red", lty = 2, cex = 0.8)

```